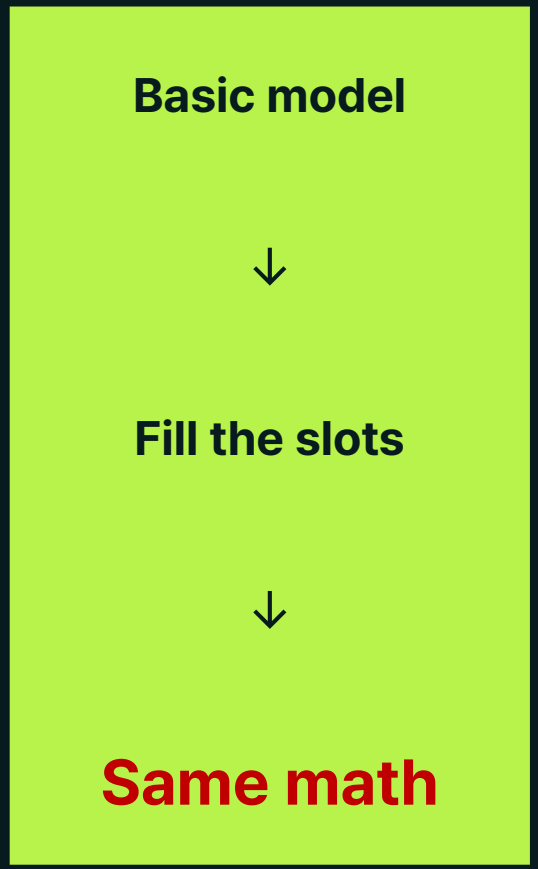
Kalman Filter Applications

Pairs Trading & Nowcasting

Same basic equations, different ingredients



Applications use the same two equations and steps

The basic model again — state and measurement equations, predict and update

System model (two equations)

$$x_k = A\,x_{k-1} + B\,u_{k-1} + w_{k-1}$$

$$z_k = H\,x_k + v_k$$

State x moves by rule A (noise Q)
Measurement z comes through H (noise R)

Repeating two steps

Step 1 Predict

$$\hat{x}_k^- = A\,\hat{x}_{k-1} + B\,u_{k-1}$$

$$P_k^- = A\,P_{k-1}A^T + Q$$

Step 2 Update

$$K_k = P_k^- H^T (H P_k^- H^T + R)^{-1}$$

$$\hat{x}_k = \hat{x}_k^- + K_k\,(z_k - H\,\hat{x}_k^-)$$

$$P_k = (I - K_k H)\,P_k^-$$

There are no new equations — only the ingredients for the slots change

An application is what goes in each slot

Ingredient table for the six slots x, A, z, H, Q, R

	Pairs trading	Nowcasting
State x	Hedge ratio β	Economic state (common factor f)
Rule A	1 (β = yesterday + wobble)	Rule moving the factor
Measurement z	EWC price Y	Employment, production, retail sales, etc.
Observation matrix H	EWA price X (changes daily)	Loading λ (one per indicator)
Noise Q · R	Wobble of β · spread noise	Factor wobble · indicator noise

Fill in one column of the table and that problem's Kalman filter is complete

01

Pairs Trading — Finding the Hedge Ratio

The hedge ratio goes in the state x slot, the other asset's price in the observation matrix H slot

The hedge ratio: shares of X sold per share of Y

Basics — pairs trading bets on the relative price of two assets

Hedge ratio β

Shares of X sold against 1 share of Y

When β changes, the scale tips

A fixed β goes stale



Even if the whole market moves, the two P&Ls offset → market neutral

Only the gap between the two prices remains: the spread Y − βX

When the relationship changes, so does β: make β the state

Changing four slots gives the pairs trading model

x → β, A → 1, z → EWC price, H → EWA price

State equation → used in Step 1 (Predict)

$$x_k = A\,x_{k-1} + B\,u_{k-1} + w_{k-1}$$

x becomes β

A = 1, B u = 0

Measurement equation → used in Step 2 (Update)

$$z_k = H\,x_k + v_k$$

z becomes Y (EWC)

H becomes X (EWA)

Pairs trading

$$\beta_k = \beta_{k-1} + w_{k-1}$$

β drifts a little each day (random walk)

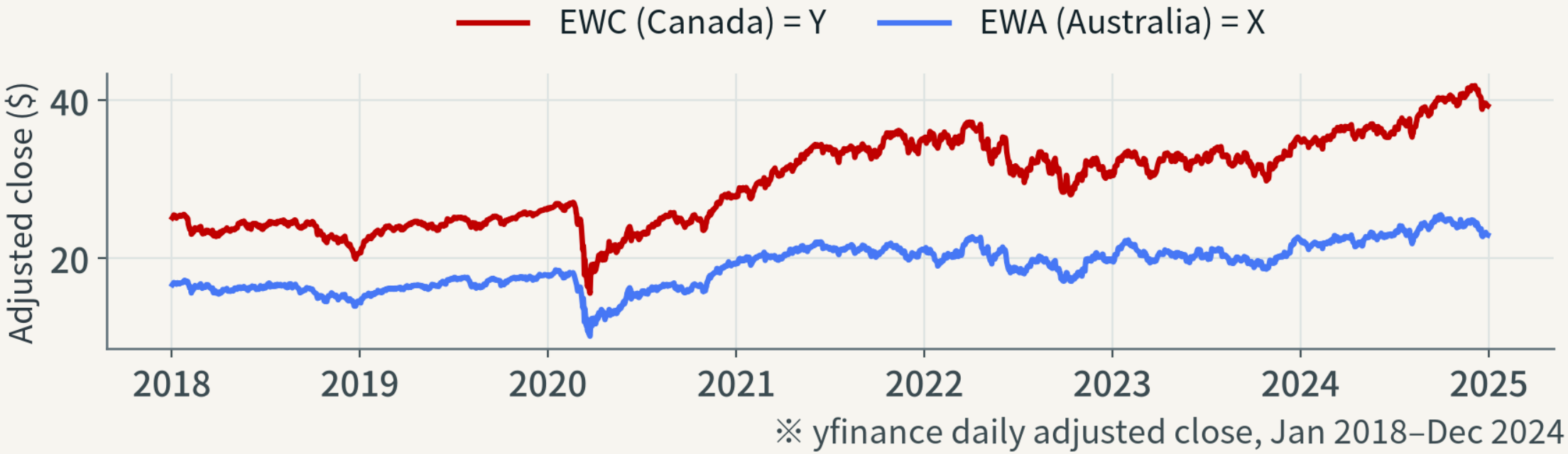
$$Y_k = X_k\,\beta_k + v_k$$

X_k in the H slot changes daily

The only difference is that a daily-changing price sits in the H slot

The example uses an Australian ETF and a Canadian ETF

EWA (Australia) and EWC (Canada) are both equities of commodity exporters — the classic pairs trading example



The two prices move together, but their ratio is not always the same

First place the two prices in the H and z slots

Step 1 prompt — downloads with yfinance, so no data file is needed

[Input] Prompt · Claude Code (Code tab)

```
In the kalman-lab folder, do the following.
1. With yfinance, download daily adjusted closes for
   EWA and EWC from 2018-01-01 to 2024-12-31
   (auto_adjust=True, Close column).
2. Save the EWA values as array X and the EWC
   values as array Y.
```

[Check] What to look for

- Daily adjusted closes for both ETFs?
- X = EWA → observation matrix H
- Y = EWC → measurement z

X goes in the H slot, Y in the z slot

The update loop reuses the five basic equations

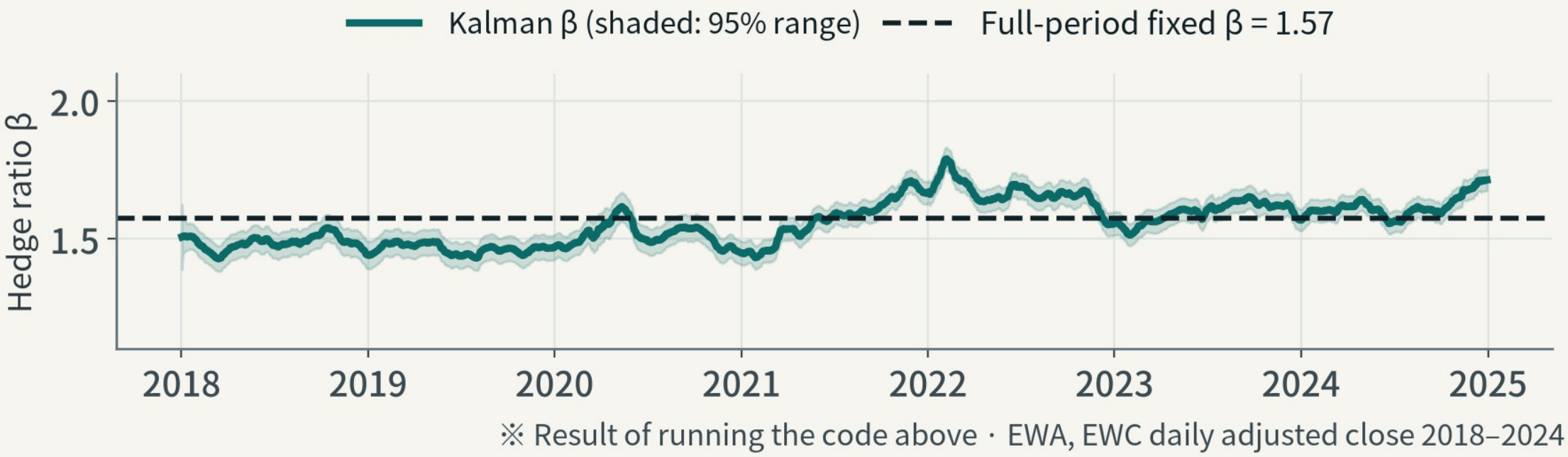
Step 2 code — the only line that differs from the basic code is 'H = Xk'

```
A, Q, R = 1.0, 1e-4, 1.0
b, P = 0.0, 1.0
for Xk, Yk in zip(X, Y):
    H = Xk
    b_pred = A * b
    P_pred = A * P * A + Q
    K = P_pred*H / (H*P_pred*H + R)
    b = b_pred + K * (Yk - H*b_pred)
    P = (1 - K*H) * P_pred
```

Rule	$A = 1, \quad Q, \quad R$
Initial	$\hat{\beta}_0 = 0, \quad P_0 = 1$
Observation matrix	$H = X_k$
①	$\hat{\beta}_k^- = A\,\hat{\beta}_{k-1}$
②	$P_k^- = A\,P_{k-1}A + Q$
③	$K_k = P_k^- X_k \,/\, (X_k P_k^- X_k + R)$
④	$\hat{\beta}_k = \hat{\beta}_k^- + K_k\,(Y_k - X_k\,\hat{\beta}_k^-)$
⑤	$P_k = (1 - K_k X_k)\,P_k^-$

The EWA·EWC hedge ratio moved between 1.43 and 1.79

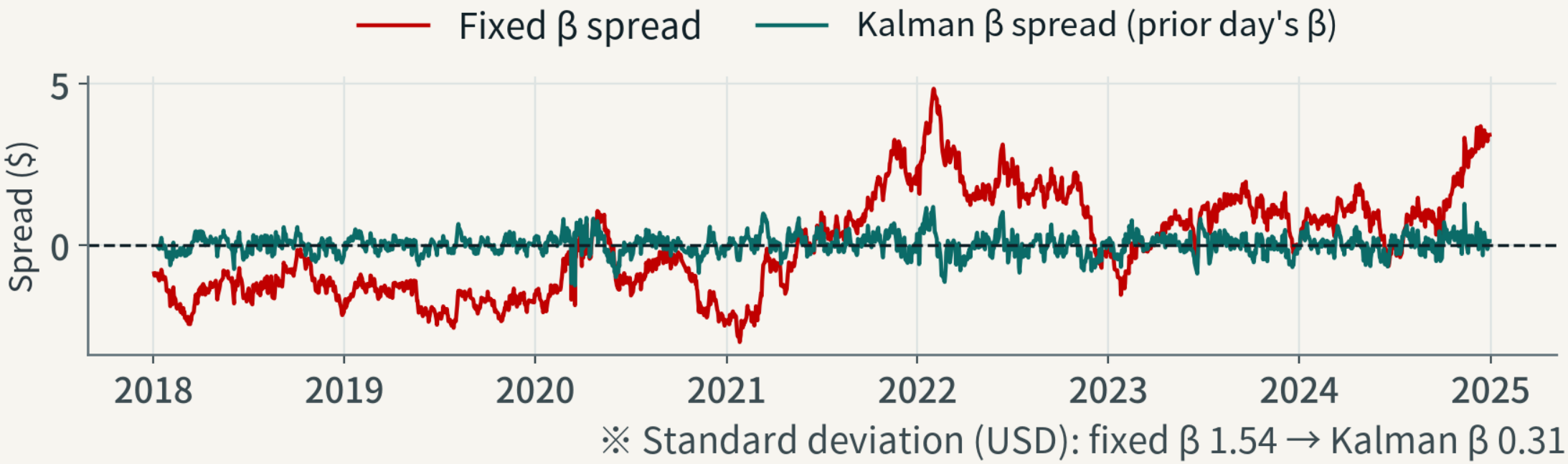
The loop above on 2018–2024 daily data — the fixed β (no-intercept regression) is the single value 1.57



Kalman β updates daily with new prices and comes with an error range

The spread built with Kalman β stays near 0

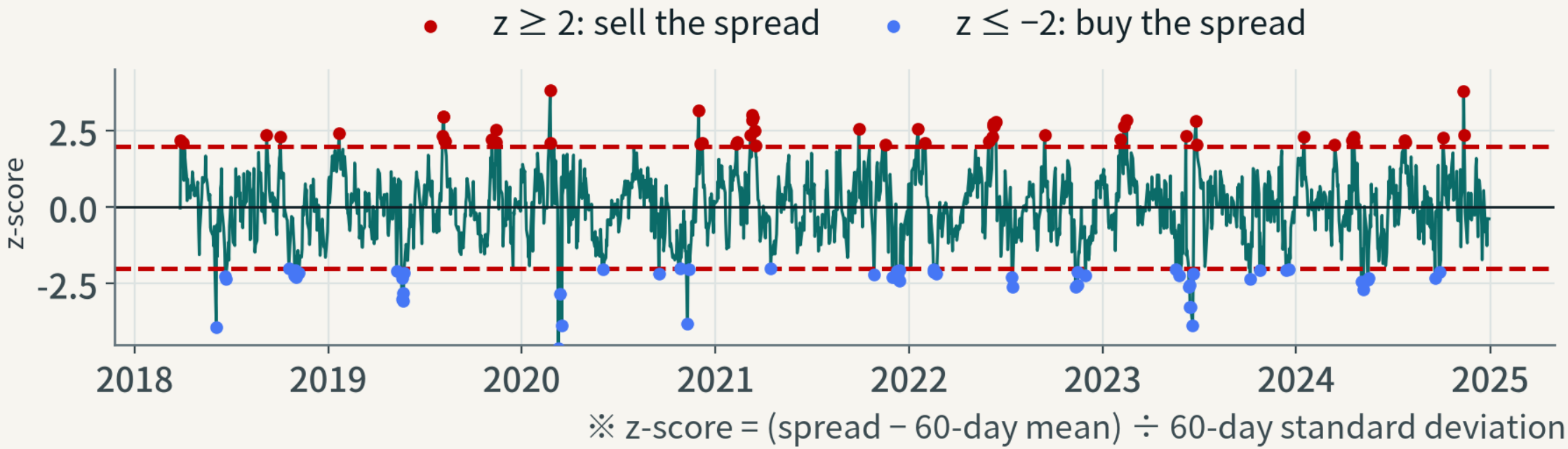
Spread = EWC price minus β times EWA price — the Kalman side uses the prior day's β known that morning



The fixed-β spread is pushed far from 0 when the relationship changes

Convert the spread to a z-score and trade at ±2

Sell the spread if z ≥ 2, buy it if z ≤ −2, and close near 0



Because the hedge ratio adapts, signals are less hit by market moves

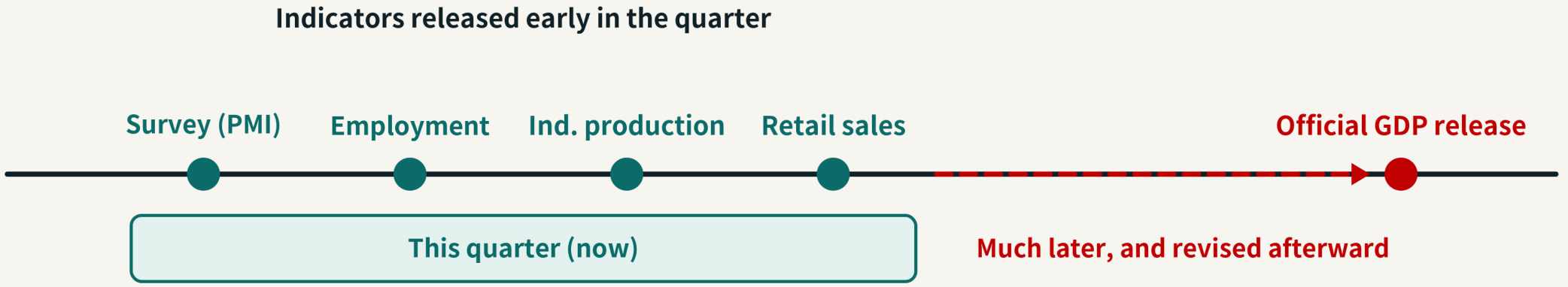
02

Nowcasting — Reading the Economy Now

The economic state goes in the state x slot, several economic indicators in the measurement z slot

Nowcasting estimates the present, not the future

Basics — official GDP comes late and is heavily revised afterward



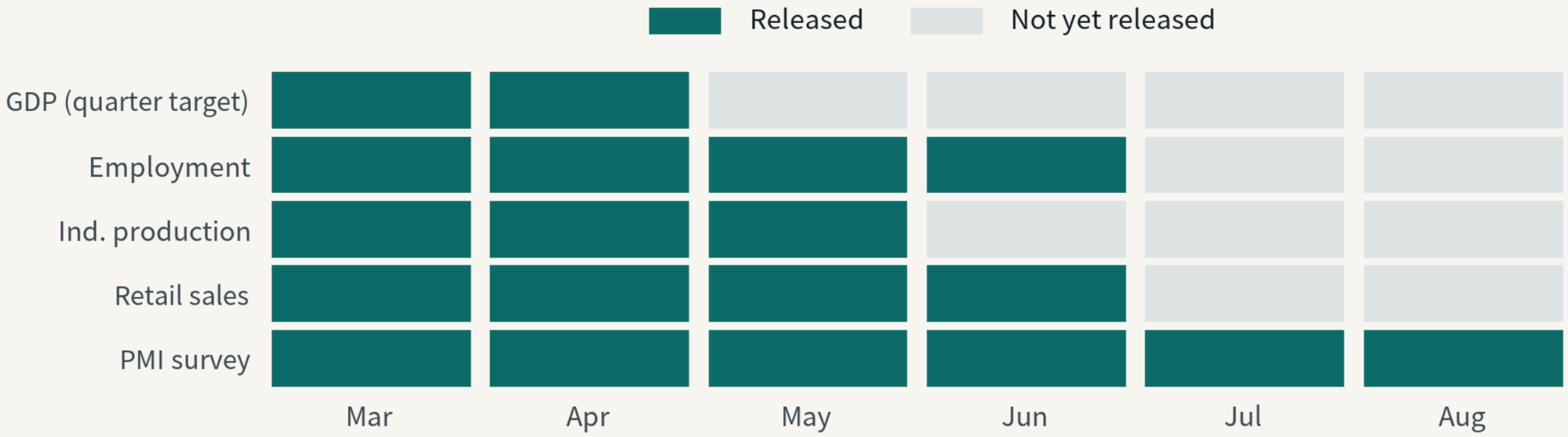
Nowcasting = estimating the 'present', not the future

Before official figures, combine early indicators to read this quarter's GDP

Giannone, Reichlin and Small (2008) set the framework; it is now a standard central bank tool

Release dates differ, so the data end is ragged

Ragged edge — surveys come fast and GDP comes last



These gaps are the reason to use the Kalman filter

Changing the slots gives a dynamic factor model

x → common factor f, z → indicators, H → loading λ

State equation → used in Step 1 (Predict)

$$x_k = A\,x_{k-1} + B\,u_{k-1} + w_{k-1}$$

x becomes factor f

B u = 0

Measurement equation → used in Step 2 (Update)

$$z_k = H\,x_k + v_k$$

z becomes indicators y

H becomes loading λ

Nowcasting (dynamic factor model)

$$f_k = A\,f_{k-1} + w_{k-1}$$

f = one hidden economic state

$$y_{i,k} = \lambda_i\,f_k + v_{i,k}$$

One row per indicator i: many measurements z

The measurement equation gets one row per indicator; same calculation

Indicators not yet released simply skip the update

With no measurement z, skip Step 2 — no interpolation or truncation is needed

	Mar	Apr	May	Jun	Jul	Aug
GDP	Update	Update	Skip	Skip	Skip	Skip
Employment	Update	Update	Update	Update	Skip	Skip
Production	Update	Update	Update	Skip	Skip	Skip
PMI	Update	Update	Update	Update	Update	Update

Rule

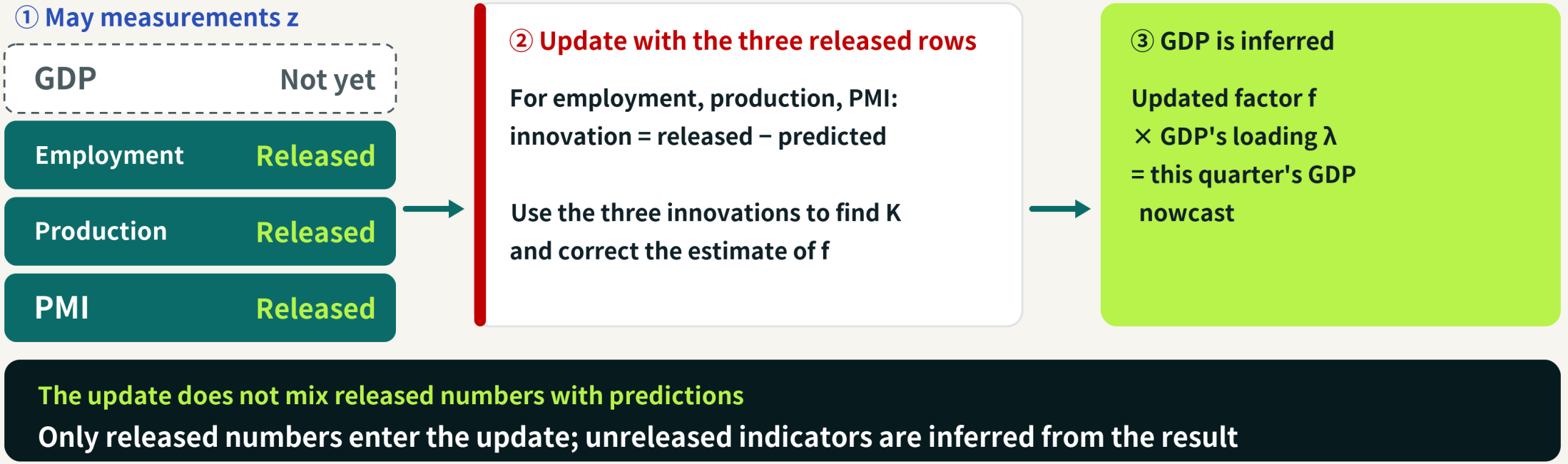
Indicators with no measurement z yet just skip the update

Step 1 (Predict) runs every month → factor f keeps moving; only Step 2 is empty

When a new release arrives, run one update with that indicator

With only some indicators, update the released rows

May example — no GDP yet; only employment, industrial production and PMI are out



Unreleased GDP is filled by the nowcast computed from the updated factor

The bracket in the update equation is the economic news

News — a released number moves the nowcast only as far as it departs from the prediction

$$\hat{x}_k = \hat{x}_k^- + K_k\,(z_k - H\,\hat{x}_k^-)$$

$(z_k - H\,\hat{x}_k^-)$ **= news: how far the released number is from the prediction**

Size of the nowcast move = K × news

As-expected number — **barely moves**

Surprise number — **moves a lot**

The nowcast revision is the gain K times the news

All at once or one at a time: the same result

Measurement z can be stacked as a vector or updated indicator by indicator

Example: predicted f = 0, P = 1 · 3 indicators (loadings 1, 0.8, 0.5 · noise 0.5, 0.5, 1 · released 0.6, 0.2, 0.4)

Method A · all at once as a vector

Stack the 3 indicators into one measurement z (a vector)
and update once

Result: f = 0.38, P = 0.22

=

Method B · one at a time

Employment +0.40 → production −0.04
→ PMI +0.02 (each one's news contribution)

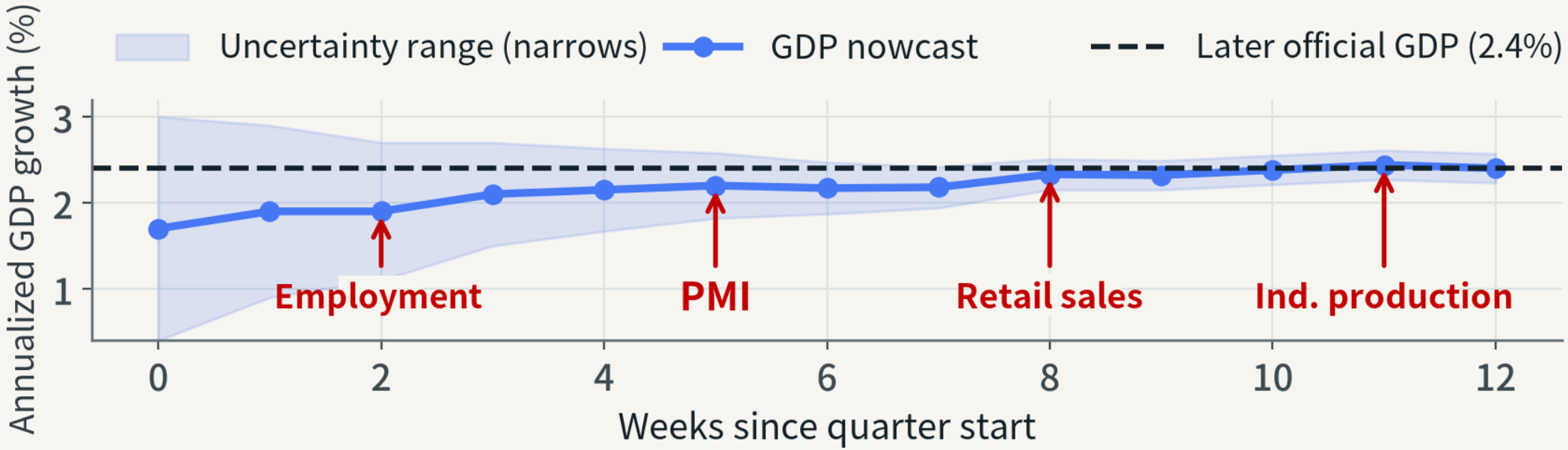
Result: f = 0.38, P = 0.22

If indicator noises are independent, both methods agree; one at a time shows each release's news contribution

News decomposition goes one at a time to get each contribution

More releases: nowcast converges, range narrows

Within the quarter, before official GDP — employment, PMI, retail sales, production released in turn



※ Redrawn from illustrative values in the original lecture figure

P shrinking with each update is the range narrowing

The basic equations are one.

Applications are the ingredients for the slots.

The calculation is repeated predict and update.

Next (Week 10): Hidden Markov Models
Hidden states with changing regimes